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Shuai Shuai

RESEARCH INTERESTS

My research focuses on the dynamics of particle-laden flows, particularly how particle inertia and preferential concentration influence particle transport and how particle–fluid interactions modify the carrier flow. I am interested in particle-induced flow modulation, vortex dynamics, and flow instability, as well as particle transport, gravitational settling, and preferential deposition in turbulent flows. My research also includes the orientation and rotational dynamics of non-spherical particles in turbulence. Using mathematical analysis and high-fidelity numerical simulations, including DNS and LES, I aim to understand the underlying physical mechanisms and develop predictive models for particle-laden flows, with particular interest in geophysical and environmental applications.

EDUCATION

Ph.D. in Mechanical Engineering Aug. 2019 – May 2024
Arizona State University, Tempe, AZ; Advisor: Mohamed Houssem Kasbaoui

M.S. in Mechanical Engineering Sept. 2015 – May 2017
Washington University in St. Louis, St. Louis, MO; Advisor: Ramesh Agarwal

B.S. in Mechanical Engineering Sept. 2010 – Aug. 2014
Shanghai Jiao Tong University, Shanghai, China

RESEARCH EXPERIENCE

Postdoctoral Researcher Aug. 2024 – Present
Cornell University, School of Civil and Environmental Engineering; Advisor: Qi Li, John D. Albertson

- Used LES and Lagrangian stochastic particle modeling to study transport, preferential concentration, and deposition of inertial particles in atmospheric boundary-layer flows with vegetation barriers.
- Quantified how vegetation-induced turbulence alters the spatial redistribution and preferential deposition of inertial particles, including changes in near-ground particle concentrations.
- Investigated how particle inertia, gravitational settling, and turbulence influence the rotational dynamics and preferential orientation of non-spherical particles in atmospheric flows.

Doctoral Researcher Aug. 2019 – May 2024
Arizona State University, School for Engineering of Matter, Transport and Energy; Advisor: Mohamed Houssem Kasbaoui

- Developed and applied two-way coupled Eulerian–Lagrangian DNS to study inertial-particle redistribution, feedback, and modulation of vortical flows.
- Derived scaling predictions for vortex decay and particle redistribution; used linear stability analysis to identify particle-induced instabilities and predict their growth rates.
- Characterized the effects of particle inertia on the merger of co-rotating vortices.

Research Assistant Aug. 2017 – Aug. 2018
Washington University in St. Louis, Department of Mechanical Engineering & Materials Science

- Developed and evaluated temperature-corrected RANS turbulence models (k–ϵ and SST k–ω) to improve predictions of high-speed compressible jets and boundary-layer flows.

PEER-REVIEWED JOURNAL ARTICLES

S. Shuai, A. Roy, and M. H. Kasbaoui, "The merger of co-rotating vortices in dusty flows," *Journal of Fluid Mechanics*, vol. 981, p. A27, 2024. DOI:10.1017/jfm.2024.67.

S. Shuai, D. Jeswin Dhas, A. Roy, and M. H. Kasbaoui, "Instability of a dusty vortex," *Journal of Fluid Mechanics*, vol. 948, p. A56, 2022. DOI:10.1017/jfm.2022.687.

S. Shuai and M. H. Kasbaoui, "Accelerated decay of a Lamb–Oseen vortex tube laden with inertial particles in Eulerian–Lagrangian simulations," *Journal of Fluid Mechanics*, vol. 936, p. A8, 2022. DOI: 10.1017/jfm.2022.50.

L. Shan, **S. Shuai**, B. Ma, Z. Du, B. Dogruoz, and D. Agonafer, "Numerical investigation of shape effect on microdroplet evaporation," *Journal of Electronic Packaging*, vol. 141, no. 4, p. 041008, 2019. DOI: 10.1115/1.4044962.

MANUSCRIPT UNDER REVIEW

S. Shuai, Q. Li, and G. Katul, "Vegetation barriers increase inertial particle deposition through preferential concentration," submitted to *Boundary-Layer Meteorology*. Preprint: 10.21203/rs.3.rs-9164043/v1.

CONFERENCE PRESENTATIONS

S. Shuai and Q. Li, "Influence of Turbulence Intensity on Non-Spherical Particle Orientation and Polarization Signals in the Atmospheric Boundary Layer," 106th AMS Annual Meeting, 2026.

S. Shuai and Q. Li, "Impact of Vegetation Barriers on Inertial Particle Transport and Preferential Deposition in Dust Storms," 25th Symposium on Boundary Layers and Turbulence, 2025.

S. Shuai and M. H. Kasbaoui, "Euler–Lagrange simulations of dust resuspension by impinging laminar and turbulent jets," APS Division of Fluid Dynamics, 2024.

S. Shuai and M. H. Kasbaoui, "Particle resuspension by an impinging turbulent jet," APS Division of Fluid Dynamics, 2023.

S. Shuai, D. Dhas, A. Roy, and M. H. Kasbaoui, "Vortex merger in semi-dilute particle-laden flow," APS Division of Fluid Dynamics, 2022.

S. Shuai, D. Dhas, A. Roy, and M. H. Kasbaoui, "Characterization of the instability of a Rankine vortex in semi-dilute dusty flows using linear stability analysis and Eulerian–Lagrangian simulations," APS Division of Fluid Dynamics, 2021.

S. Shuai, D. Dhas, A. Roy, and M. H. Kasbaoui, "Stability analysis of vortex flow with dispersed microdroplets," ASME FEDSM, 2021.

S. Shuai, A. Roy, and M. H. Kasbaoui, "Motion and instability properties of Lamb–Oseen vortex in particle-laden flows," APS Division of Fluid Dynamics, 2020.

S. Shuai and M. H. Kasbaoui, "Stability of a Lamb–Oseen vortex in two-way coupled particle-laden flows," APS Division of Fluid Dynamics, 2019.

PEER-REVIEWED CONFERENCE PROCEEDINGS

S. Shuai and R. K. Agarwal, "A New Improved One-Equation Turbulence Model Based on k-kL Closure," AIAA SciTech 2020 Forum, AIAA 2020-1075. DOI: 10.2514/6.2020-1075.

S. Shuai and R. K. Agarwal, "Modification of Wray–Agarwal Transition Model for Computing Rough Wall Flows," AIAA SciTech 2019 Forum, AIAA 2019-0333. DOI: 10.2514/6.2019-0333.

S. Shuai, Z. Du, B. Ma, L. Shan, B. Dogruoz, and D. Agonafer, "Numerical Investigation of Shape Effect on Microdroplet Evaporation," ASME InterPACK 2018, IPACK2018-8311. DOI: 10.1115/IPACK2018-8311.

S. Shuai, T. Wray, and R. K. Agarwal, "Turbulence Modeling of High Speed Compressible Flows," AIAA SciTech Forum – 55th AIAA Aerospace Sciences Meeting, AIAA 2017-0317. DOI: 10.2514/6.2017-0317.

DISSERTATIONS AND THESES

S. Shuai, "Understanding the Mechanism of Vortex Flow Modulation by Inertial Particles," Ph.D. Dissertation, Arizona State University, 2024. https://keep.lib.asu.edu/items/193358

S. Shuai, "Temperature Corrected Turbulence Models for High Speed Compressible Flows," M.S. Thesis, Washington University in St. Louis, 2017. DOI: 10.7936/K71V5DDD.

PROFESSIONAL SERVICE

Reviewer for *Journal of Fluid Mechanics* (4 manuscripts).

Reviewer for *Journal of Advances in Modeling Earth Systems* (1 manuscript).

Contributed to the organization of the Global Network of Networks of Integrated Urban Services (GNNIUS) Workshop, Cornell University, 2024. gnnius.engineering.cornell.edu

INVITED TALKS

"The Vortex Flow Modulation by Inertial Particles in Two-Way Coupling," Environmental Fluid Mechanics & Hydrology Seminar, Cornell University, Dec. 3, 2024.

TEACHING EXPERIENCE

Teaching Assistant and Guest Lecturer, *CEE 3310: Fluid Mechanics*, Cornell University, Fall 2024.

Guest Lecturer, *MAE 574: Multiphase Flow*, Arizona State University, Fall 2023.

TECHNICAL SKILLS

Numerical Methods: Direct Numerical Simulation (DNS), Large-Eddy Simulation (LES), two-way coupled Eulerian–Lagrangian modeling, particle Lagrangian Stochastic Modeling (LSM).

Mathematical Analysis: Linear Stability Analysis (LSA).

Programming & Computing: Python, MATLAB, Fortran, Message Passing Interface (MPI), high-performance computing (HPC).

HONORS & AWARDS

Graduate College Completion Fellowship, Arizona State University, 2023 (full tuition and stipend support).

Top Paper in Track 4, ASME 2018 InterPACK, 2018.